

Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) for CFD

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August 2026

Background and research challenge

- Physics-Informed Neural Networks (PINNs) offer mesh-free PDE solving for Computational Fluid Dynamics (CFD) by enforcing physical laws directly in loss functions.
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- Parameter explosion increases memory footprints, slows down evaluation times, and demands extensive computational resources.

Proposed Solution: QAPINNs

- To overcome these limitations, Quantum-Assisted Physics-Informed Neural Networks (QAPINNs) embed a Variational Quantum Circuit (VQC) into the neural network architecture.
- Replacing classical layer with Quantum layer reduces the number of trainable parameters significantly. Within a QAPINN, entanglement enables interactions between spatial and temporal dimensions in Hilbert space, allowing the network to coordinate spatio-temporal updates during optimization.

Mathematical Framework & Benchmark PDEs

Burger's equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad x \in [-1, 1], \quad t \in [0, 1]$$

- Initial Condition (IC): $u(x, 0) = -\sin(\pi x)$
- Boundary Conditions (BC): $u(-1, t) = u(1, t) = 0$
- Kinematic Viscosity: $\nu = \frac{0.01}{\pi}$

QAPINN Architecture Design Choice

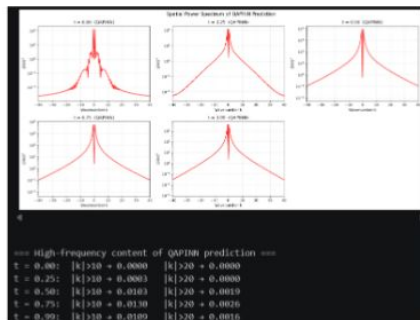
- In QAPINN, third classical hidden layer will be replaced by quantum layer. Encoding will be done by angle embedding function in pennylane.
- We used Adam optimizer with a learning rate of 0.005. A learning rate scheduler, ReduceLROnPlateau, is employed to adjust the learning rate when the loss plateaus dynamically. Specifically, the scheduler reduces the learning rate by a factor of 0.9 if no improvement is observed for 1000 consecutive epochs.
- During training, Pauli-Z expectation measurements are performed on each qubit to extract features from the quantum state. The number of epochs used for all situations is between 12000 to 15000. Neural network is trained with data set of size 2500.

Performance & Parameter Efficiency Results(Burger's equation)

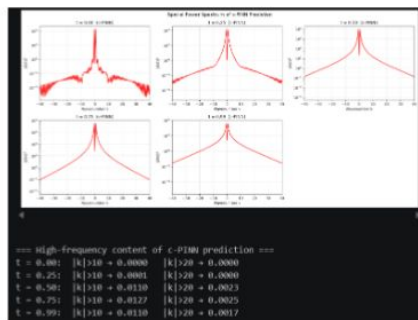
	Parameters	L_2 error	Final loss
c-PINN	7851	1.9387e-01	0.0005
3 Qubits	581	1.9429e-01	0.01307
4 Qubits	691	8.1445e-02	0.0055
5 Qubits	801	6.9634e-02	0.0045

- We only trained neural network with data set of size 2500. If we train the c-PINN with more more points(25600), it is performing better than QAPINN. But QAPINN is able to achieve this result with only 2500 uniformly distributed points.
- when c-PINN is trained with data set of size 25600, L_2 error is 3.8254e-02 and final loss is 0.0004.

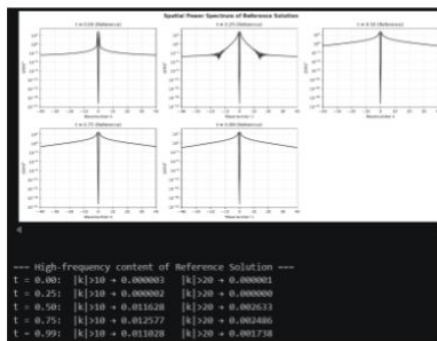
Spectral Analysis



(a) Analyzed using 4 qubits



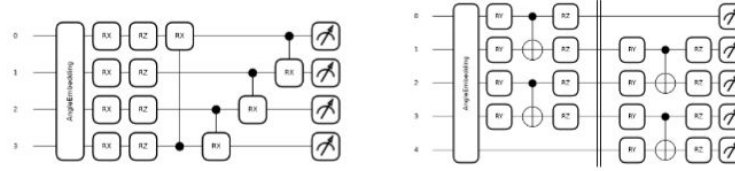
(b) spectral analysis for c-PINN solution



(c) spectral analysis for reference solution

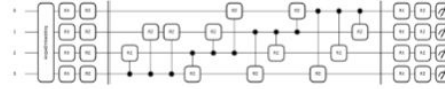
- we performed a spatial Fast Fourier Transform (FFT) on the predicted solution field $u(x, t)$ across five discrete time steps.
- While both network architectures successfully avoid spectral bias, QAPINN demonstrates tighter alignment with the reference benchmark at peak high-frequency energy states ($t=0.75$, $|k| > 20$: 0.0026 for QAPINN vs. 0.0025 for c-PINN vs. 0.002486 for Reference).

Ansatz Selection



(a) cascade

(b) Alternate

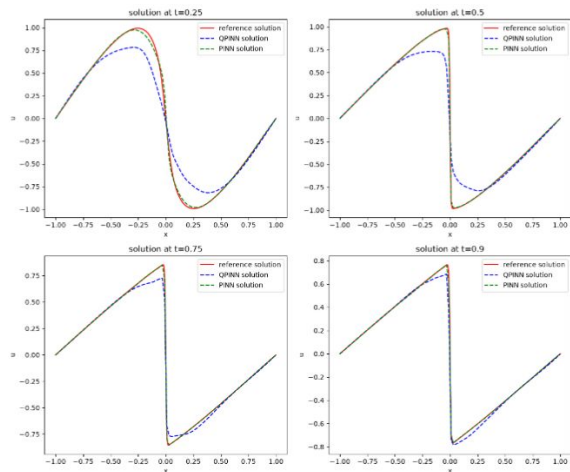


(c) cross mesh

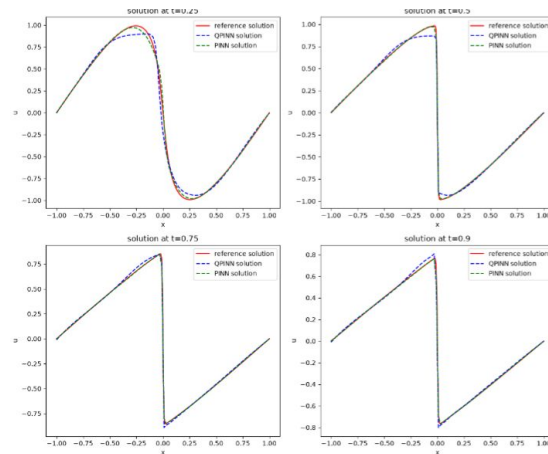
Cascade	cross mesh	Alternate	strongly entangled layer	Basic entangled layer
0.0045	0.0187	0.0195	0.0133	0.0140

Table 2: Total loss after the completion of training of neural network

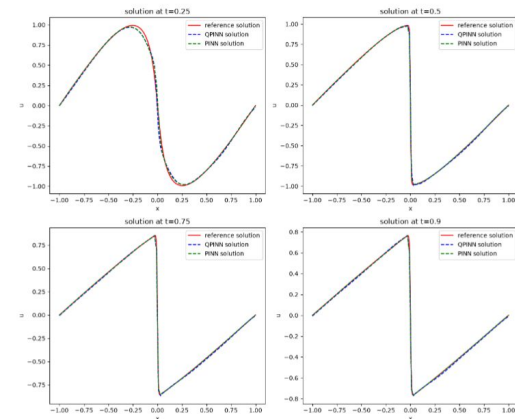
visualizing shock wave at different time intervals



(a) using 3 Qubits

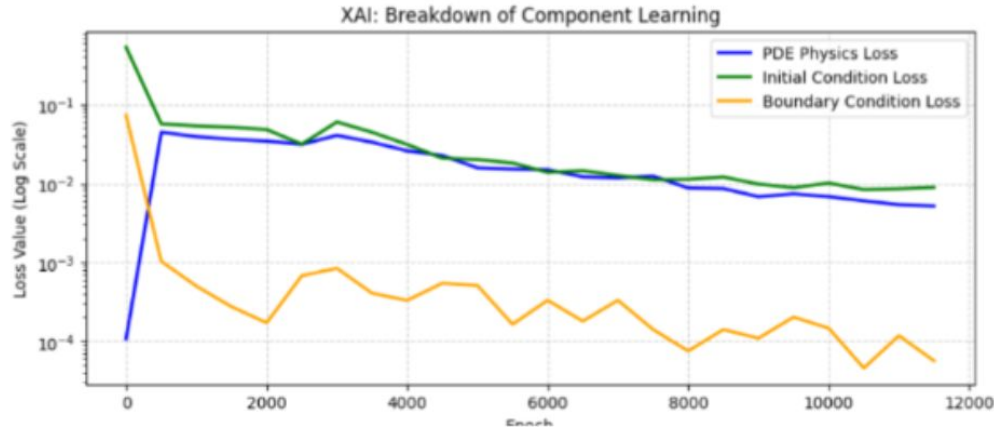


(b) using 4 Qubits



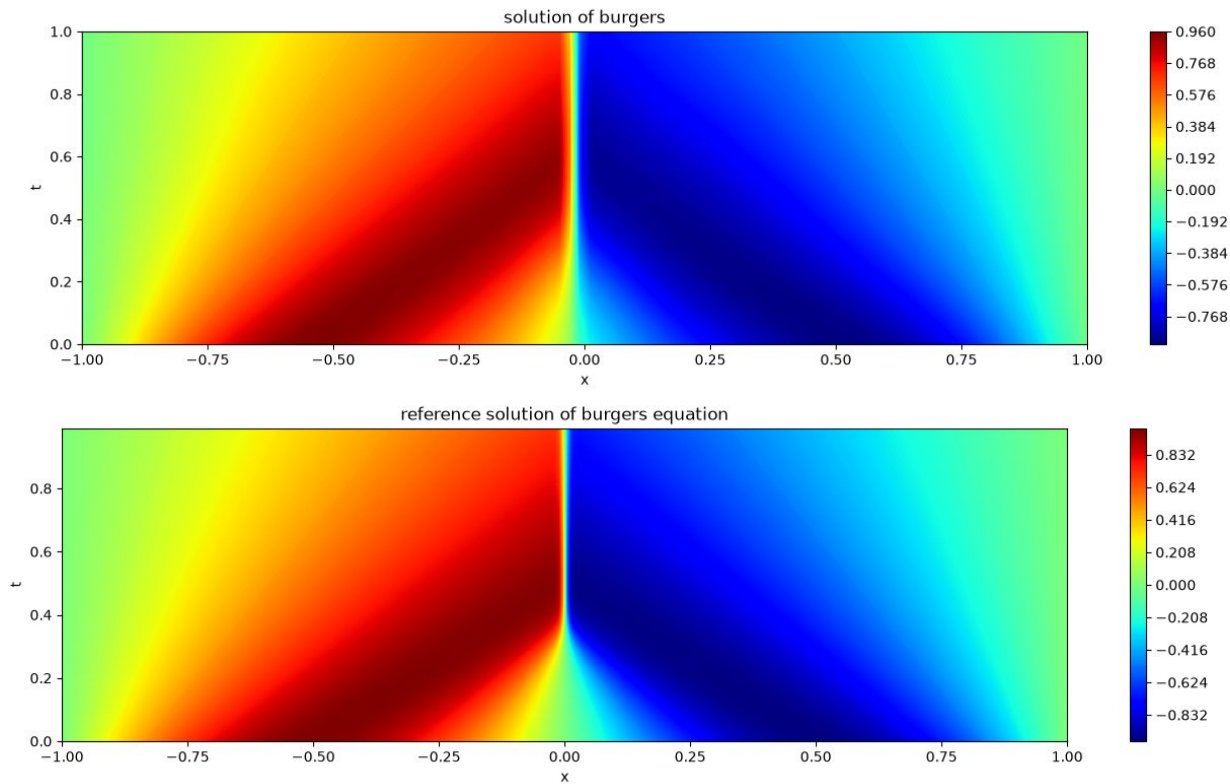
(c) using 5 Qubits

Explainable AI (XAI) Diagnostics



Explicitly demonstrate how the hybrid network learns, we isolated the composite loss function into its three constituent parts: the PDE residual, the initial conditions, and the boundary conditions.

Comparing contour plots



Conclusion

- **Parameter Reduction:** Over 89% reduction in trainable parameters compared to classical baselines.
- **Data Efficiency:** High pointwise physical accuracy on sparse datasets (2,500 points) where c-PINNs struggle.

References

- When Do Extended Physics-Informed Neural Networks (XPINNs) Improve Generalization?<https://arxiv.org/abs/2109.09444>
- Physics Informed Deep Learning (Part I): Data-driven Solutions of Non-linear Partial Differential Equations.<https://arxiv.org/abs/1711.10561>
- Benchmarking Quantum-Assisted PINN (QA-PINN) for Computational Fluid Dynamics.
- The effect of data encoding on the expressive power of variational quantum machine learning models.<https://arxiv.org/abs/2008.08605>
- QCPINN: Quantum-Classical Physics-Informed Neural Networks for Solving PDEs.<https://arxiv.org/abs/2503.16678>
- Turning quantum nodes into Torch Layers(pennyLane tutorial)